

The Scientific Method and Zeller's Formula Test a 3,400-Year-Old Calendar's Exact Mathematical Dating Claim

Math Lessons from the Continuous Jubilee Calendar

Math Lessons · Continuous Jubilee Calendar

"It is written in the language of mathematics." — Galileo Galilei, on the universe, in The Assayer (1623)

Lesson 0: Where Are We Now?

Can you figure out what day of the week your birthday will fall on in **2046**, or what day it was **2,000 years ago**, with nothing but a pencil? And can you find where any year, past or future, sits on a calendar that began **more than 3,400 years ago** and is **still being counted today**?

By the end of these lessons, you'll be able to do both. You'll learn two short formulas: one that finds any year's place on an ancient calendar, and a bonus one that finds the day of the week for any date. All you need is subtraction, multiplication, division with remainders, and careful thinking.

A calendar that counts years in sevens

You already know one calendar that counts in sevens: the week. Many cultures have followed seven-day weeks for thousands of years, with a regular day of rest.

Now imagine counting **years** the same way. An ancient law code, the book of **Leviticus** (part of the Hebrew Bible, written in the ancient Near East), describes exactly that kind of calendar:

"Six years thou shalt sow thy field... But in the seventh year shall be a sabbath of rest unto the land" (Leviticus 25:3-4).

So every seventh year was a **rest year**, when no crops were planted and the land was left alone. (It's also called a **sabbatical year**. Its Hebrew name is ***shemitah***.)

Then comes a bigger pattern:

"Seven sabbaths of years... seven times seven years; and the space of the seven sabbaths of years shall be unto thee forty and nine years" (Leviticus 25:8).

Seven weeks of years is $7 \times 7 = 49$ **years**. After each 49-year cycle came a special year called the **Jubilee**, a year when debts were cancelled, servants were freed, and families got back land they had sold.

You may have seen the Jubilee law before. The **Liberty Bell** in Philadelphia, one of the most famous symbols of American independence, is inscribed with a line from it: "Proclaim LIBERTY throughout all the Land unto all the Inhabitants thereof" (Leviticus 25:10).

So this calendar has three parts:

- **a week of years:** 7 years, with the 7th year a rest year;
- **a cycle:** 7 weeks of years = 49 years;
- **the Jubilee:** the year that starts each new cycle.

The **Continuous Jubilee Calendar (CJC)** counts these years without ever stopping, from its starting point more than 3,400 years ago to today.

It's still being counted

The seven-year count from Leviticus is **still kept today**. On the Jewish calendar, the years 2021–2022 were a *shemittah*, a rest year. In Lesson 1 you'll check that year with the CJC formula yourself.

Why calendar changes don't break it

People have changed calendars many times:

- The **BC/AD** system for numbering years wasn't invented until about **AD 525**, by a monk named Dionysius. It has one strange feature you'll meet in Lesson 2: there is **no year zero**.
- In **1582**, much of Europe switched from the old **Julian** calendar to the **Gregorian** calendar we use today, and **ten days** were skipped to fix a drift in the seasons.

Neither change breaks the CJC formula. The CJC counts **years**, not days: skipping ten days moved the dates, but it didn't add or remove a single year. As long as we handle the missing year zero carefully, the count stays exact.

Precise versus accurate

The formula is **exact**: put in any year and you'll always get the same answer, with no rounding and no guessing, and anyone can check it by hand. In math and science, that's called being **precise**.

But being precise isn't the same as being right. Being right is called being **accurate**. Picture a dartboard:

- darts tightly clustered **but far from the center** are precise, but not accurate;
- darts tightly clustered **in the center** are precise **and** accurate.

The CJC formula always hits the same spot, so it's perfectly precise. Is that spot the bullseye, meaning do its answers match real history? That's a question you can **test**, and in Lesson 5, you will.

Lesson 1: The Formula

The CJC count starts at **1437 BC**. That year is **number 1** on the count, and every year after it gets the next number.

Step 1: Find B (the count number)

- For a **BC** year: **$B = 1438 - \text{the year}$**
- For an **AD** year: **$B = \text{the year} + 1437$**

Quick check: 1437 BC $\rightarrow 1438 - 1437 = 1$. The first year is number 1. ✓

Step 2: Find the Cycle (which 49-year cycle)

Subtract 1 from B and divide by 49. **Drop the remainder**, then add 1.

$$\text{Cycle} = (B - 1) \div 49, \text{ remainder dropped, } + 1$$

Step 3: Find Y (the year inside the cycle, 1 to 49)

$$Y = B - (\text{Cycle} - 1) \times 49$$

Step 4: Find the Position (the year inside its week, 1 to 7)

Divide Y by 7 and look at the **remainder**:

- **remainder 1 to 6:** that's the position;
- **remainder 0:** position 7, a **rest year**.

Special case: if **Y = 1**, the year is a **Jubilee** (and position 1, the first year of a new week).

Worked example: where are we now? (AD 2026)

- **$B = 2026 + 1437 = 3,463$**
- **Cycle:** $3,462 \div 49 = 70$, remainder 32. Drop the remainder: $70 + 1 = \text{Cycle } 71$
- **$Y = 3,463 - (70 \times 49) = 3,463 - 3,430 = 33$**
- **Position:** $33 \div 7 = 4$, remainder 5 $\rightarrow$ **Position 5**

2026 is Cycle 71, Year 33, Position 5: the fifth year of a seven-year week. The next rest year is two years away.

Try it yourself

- **1.** Check the Jewish rest year: **AD 2021**. Is it a rest year on the CJC?
- **2.** When is the **next rest year**? Try **AD 2028**.
- **3.** When is the **next Jubilee**? Try **AD 2043**.

Lesson 2: The Missing Year Zero

How many years are there from **2 BC** to **AD 2**? Most people say 4. Look at the number line:

2 BC → 1 BC → AD 1 → AD 2

There is **no year zero**. The year after 1 BC is AD 1. So from 2 BC to AD 2 there are only **3 steps**, not 4.

The rule: when a count of years **crosses from BC into AD**, subtract as usual, then **add 1**.

Worked example: a 483-year prophecy

The book of **Daniel**, another ancient text in the Hebrew Bible, contains a famous prediction measured in “weeks” of years. It says that from an order “to restore and to build Jerusalem” until a promised leader, called “the Messiah the Prince,” would be “seven weeks, and threescore and two weeks” (Daniel 9:25). “Threescore” means 60, so that’s $7 + 62 =$ **69 weeks of years**:

$$69 \times 7 = \mathbf{483 \text{ years}}$$

The CJC starts this count at an order from the Persian king **Artaxerxes** in **457 BC**. Where do 483 years land?

- $483 - 457 = 26$
- We crossed from BC into AD, so add 1: **AD 27**

AD 27 is the year the CJC gives for the start of the public career of **Jesus of Nazareth**, whom his followers identified as that promised leader. Check it with the formula: $\text{AD } 27 \rightarrow B = 1,464 \rightarrow$ **Cycle 30, Year 43, Position 1**, the **first year of a new week of years**.

Try it yourself: test other starting points

Other Persian kings also gave orders about Jerusalem. Add 483 years to each. Careful: only some of these cross into AD!

- **4.** 538 BC (King Cyrus)
- **5.** 520 BC (King Darius)
- **6.** 444 BC (Artaxerxes again)

Worked example: seventy weeks = ten Jubilee cycles

Daniel's full prophecy begins, "Seventy weeks are determined" (Daniel 9:24). That's **70 weeks of years**:

$$70 \times 7 = \mathbf{490 \text{ years}} = 10 \times 49, \text{ exactly } \mathbf{ten \text{ Jubilee cycles}}$$

First, check where the count starts. $\mathbf{457 \text{ BC}} \rightarrow B = 1438 - 457 = \mathbf{981} \rightarrow 980 \div 49 = 20$, remainder 0 $\rightarrow$ **Cycle 21** $\rightarrow Y = 981 - (20 \times 49) = 981 - 980 = \mathbf{1}$. The order of 457 BC falls in a **Jubilee** year!

Now go forward 490 years:

- $490 - 457 = 33$
- We crossed from BC into AD, so add 1: **AD 34**
- Check it: $\text{AD } 34 \rightarrow B = 1,471 \rightarrow 1,470 \div 49 = 30$, remainder 0 $\rightarrow$ **Cycle 31** $\rightarrow Y = \mathbf{1}$. **Another Jubilee.**

Daniel's seventy weeks run **from one Jubilee to another, exactly ten cycles later**. Every number fits perfectly, because 490 is exactly 10×49 .

- **7.** The seventy weeks end with **one final week** of seven years, which starts in AD 27. Which seven years does the final week cover? Which year is its **middle** (Position 4)?

Lesson 3: Counting Days

The middle of the week

- **8.** Historians most often date the crucifixion of Jesus to **AD 30** or AD 33, and the CJC uses **AD 30**. Find AD 30's position in the seven-year week.

You should get **Position 4**. In a week of 7, position 4 is the exact middle: three years before it and three after. Daniel's prophecy says the promised leader would be "cut off... in the midst of the week" (Daniel 9:26-27).

"The third day": inclusive counting

The Gospels (four ancient accounts of Jesus' life) report that he was crucified on a **Friday**, which in AD 30 was April 7, and that his followers said they saw him

alive again “the third day,” on **Sunday**. (In the Bonus Lesson, you’ll prove with your own arithmetic that April 7, AD 30 really was a Friday.)

Wait: Friday to Sunday is only two days later! The answer is **inclusive counting**, which people in the ancient world used all the time: **the starting day counts as day 1**.

Friday	Saturday	Sunday
day 1	day 2	day 3

Forty days and fifty days

The book of Acts reports that Jesus’ followers saw him over a period of “forty days” (Acts 1:3). The ancient harvest festival of **Pentecost** was counted as **fifty days**: “number fifty days” (Leviticus 23:16).

Count inclusively, starting with **Sunday, April 9, AD 30** as day 1. (Hint: April has 30 days.)

- **9.** What date is **day 40**? What day of the week?
- **10.** What date is **day 50**? What day of the week?

Notice something? 50 days is 7 weeks plus 1 day, so day 50 is both the end of the count and the start of a new week. That’s the same pattern as the Jubilee: after $7 \times 7 = 49$ years, the 50th year is also year 1 of the next cycle.

Forty years

In the Hebrew Bible, a **generation** is often counted as forty years (“Forty years long was I grieved with this generation,” Psalm 95:10). The Gospels report that Jesus warned that the great **Temple** in Jerusalem would be destroyed within “this generation” (Matthew 23:36-38).

- **11.** Add 40 years to AD 30. (No BC-to-AD crossing, so no extra step!) The Roman army destroyed the Temple in that very year.

Lesson 4: Multi-Step Problems

A rescue in a Jubilee

Around **702 BC**, the army of the **Assyrian** Empire surrounded Jerusalem, but the city survived.

- **12.** Find 702 BC on the CJC. What kind of year is it?

When the captivity began

In **605 BC**, the army of **Babylon** carried young men from Jerusalem away as captives. (A **captivity** is when a people are taken from their homeland.) The prophet **Jeremiah** said the captivity would last “seventy years” (Jeremiah 25:11).

- **13.** Find 605 BC on the CJC. What is Y? What kind of year is it?
- **14.** Subtract 70 from 605. Where does that year land?

Something to notice: $70 = 10 \times 7$. So if you start on a rest year and count 70 years, you’ll always land on another rest year. That’s not magic; it’s multiplication.

Division with remainders: 430 years

The book of **Ezekiel** describes a sign that counts **430 years** (Ezekiel 4:5-6). How many special years (rest years plus Jubilees) fit into 430 years on the CJC?

- **Divide:** $430 \div 49 = 8$, remainder **38**. So 430 years is 8 full cycles plus 38 extra years.
- **Each full cycle** has 7 rest years + 1 Jubilee = **8 special years**. $8 \times 8 = 64$.
- **The 38 extra years** hold a Jubilee (year 1) and rest years at years 7, 14, 21, 28, and 35: **6 more**.
- **Total:** $64 + 6 = 70$

That’s the same **seventy** that ancient Hebrew texts give for the length of the Babylonian captivity.

Not every year is special

A good scientist checks the misses too.

- **15.** Find **586 BC** (the year Babylon destroyed Jerusalem) and **538 BC** (the year the Persian king Cyrus let the captives go home). Are they rest years or Jubilees?

You should find they’re **ordinary years**. Big events don’t always land on special years, and a formula that claimed they did would be suspicious.

Lesson 5: Checking Against History

This is the most important lesson, because it’s where we test the formula.

Some old records say a certain year was a rest year, or a certain year in the seven-year cycle. They were written long ago by people who never heard of the CJC. If the formula is **accurate**, not just precise, it should **predict** what those records say.

This is the scientific method.***1. Question:*** Does the formula match the real seven-year count?***2. Hypothesis:*** The formula puts the rest years in the right place.***3. Prediction:*** If the hypothesis is right, years that old records call rest years will come out as rest years in the formula.***4. Test:*** Calculate those years, then compare the results with records written by people who never heard of the CJC.***5. Conclusion:*** State what the results show, and how strong the evidence is.***6. Share and repeat:*** Anyone can redo the calculations and check them.**Testing the rest years**

One thing to know first: the ancient Jewish year started in the fall, so each of their years covers parts of two of ours, like a school year (“2025–26”). When a record says “164/163 BC,” use the **first** year in the formula.

Year	What the old record says	Your job
164/163 BC	The ancient history 1 Maccabees says that defenders of a town under siege ran out of food, “it being a year of rest to the land” (1 Maccabees 6:49).	16. Rest year?
38/37 BC	The ancient historian Josephus wrote that when King Herod besieged Jerusalem, it was a rest year.	17. Rest year?
AD 358	At Zoar, an ancient town by the Dead Sea (in modern Jordan), archaeologists found gravestones that give the year of death as a year in the seven-year cycle. One from December AD 358 says “the third year of the sabbatical cycle.”	18. What position?
2021/22	The Jewish calendar kept it as a rest year.	(You checked this in Lesson 1!)

How strong is the evidence? A lesson in probability

All of these records follow the **same** seven-year rhythm. Once the formula matches one, it will match the others too. So they really count as **one test**, not four.

What are the chances of passing that test by luck? There are 7 different ways to line up a seven-year rhythm, and only 1 of them matches the records. A formula that guessed at random would pass **1 time out of 7**, about a 14% chance.

- **19.** If you rolled a fair seven-sided die, what’s the probability of rolling a 7? How is that like this test?

Passing a 1-in-7 test is real evidence, but not proof. Good scientists always say exactly how strong their evidence is, and stating it is part of step 5.

Testing the famous dates

Daniel's prophecy also makes predictions that can be checked against dates historians already use:

Prediction from the ancient text	What we calculate	What historians say	Result
The promised leader appears 483 years after the order of 457 BC (Daniel 9:25)	$457 \text{ BC} + 483 \text{ years} = \text{AD } 27$ (Lesson 2)	The Gospel of Luke dates the start of Jesus' public career to "the fifteenth year" of the emperor Tiberius (Luke 3:1). Depending on how Tiberius's reign is counted, historians place that around AD 27 to 29.	✓ It fits the range
He is "cut off" in "the midst of the week" (Daniel 9:26-27), meaning Position 4	AD 30 → Position 4, the exact middle. AD 33 → Position 7, the end of the week.	Historians' two leading dates for the crucifixion are AD 30 and AD 33.	✓ AD 30 fits; AD 33 doesn't
The Gospels: crucified on a Friday, at Passover	Zeller's formula (Bonus Lesson): April 7, AD 30 → Friday	Astronomers calculate that the Passover full moon fell that week.	✓
Seen alive "the third day"	Friday = day 1, Saturday = day 2 → Sunday, April 9 = day 3	The Gospels report the first day of the week (Sunday).	✓

Here's the interesting part. Both of historians' candidate dates, April 7, AD 30 and April 3, AD 33, turn out to be **Fridays**. So the day of the week can't tell them apart. But the CJC can: only AD 30 lands in "the midst of the week." One test that can't decide, and one that can, is exactly how scientists narrow down possibilities.

- **20.** Use the Julian formula in the Bonus Lesson to check that **April 3, AD 33** was also a Friday. Then find **AD 33** on the CJC. What position is it?

Lesson 6: Big Numbers

Seventy cycles

In many ancient texts, **seventy** marks something being **complete**: seventy years of captivity, seventy weeks in Daniel's prophecy.

- **21.** How many years are **70 cycles** of 49 years?
- **22.** Starting from 1437 BC, where do that many years land? (Careful: you cross from BC into AD!)

- **23.** Check your answer with the formula. What kind of year is it?

Counting backward by 49

Every Jubilee is 49 years after the one before. Once you know one Jubilee, you can find others by adding or subtracting 49.

- **24.** Subtract 49 from your answer to problem 22. What famous year is it? Check it with the formula.

That year, **1945**, is when World War II ended and the prisoners of the Nazi concentration camps were set free. Remember the Jubilee's words on the Liberty Bell: "Proclaim LIBERTY throughout all the Land."

1,260 years

The books of Daniel and Revelation both describe a period of **1,260 days** (Revelation 12:6). The same span is also written as "a time, and times, and half a time" (Revelation 12:14; see Daniel 7:25).

Why would anyone count those days as **years**? Because the ancient texts themselves give that rule for prophecies. Two passages spell it out:

- "each day for a year" (Numbers 14:34), where forty days of scouting become forty years of wandering;
- "I have appointed thee each day for a year" (Ezekiel 4:6).

You've already used this idea. In Lesson 2, Daniel's "weeks" were counted as **weeks of years**, 7 years each, not 7 days, and that's how 69 weeks became 483 years and landed on AD 27. Readers who apply the same "day for a year" rule to the 1,260 days get **1,260 years**.

Many of those readers date the period from **AD 538**, when a decree of the Roman emperor Justinian giving the bishop of Rome authority over the churches could first take effect in the city, to **1798**, when a French army entered Rome and took the pope prisoner.

- **25.** Add 1,260 to 538. Then find **538** and your answer on the CJC.

You should find that 538 is **position 1**, the start of a week of years, and the end year is a **Jubilee**. Here's why: $1,260 = 25 \times 49 + 35$, and 538 is Year 15 of its cycle, so $15 + 35 = 50$, which is Year 1 of the next cycle.

A warning about "finding" patterns

Problems 23–25 are **observations**, not tests. No old record said "1945 will be a Jubilee." Search through enough years and you'll always find some famous event on a special year. That's why Lessons 4 and 5 matter so much: they include the

misses and the **real tests**. A good mathematician enjoys a surprising pattern, but trusts only a pattern that has been tested.

Where the count is anchored

The Jubilee count isn't simply picked to fit famous years. The ancient texts give it **checkpoints**.

When the Assyrian army surrounded Jerusalem around 702–701 BC, the prophet **Isaiah** described two years in a row without planting: “Ye shall eat this year such as groweth of itself; and the second year that which springeth of the same: and in the third year sow ye, and reap” (Isaiah 37:30). Two years without planting, back to back, is exactly what a **rest year followed by a Jubilee** would produce. And that's where the formula puts them: **703 BC is Year 49 (a rest year), and 702 BC is Year 1 (a Jubilee)**.

Daniel's prophecy is another checkpoint: its seventy weeks are exactly ten Jubilee cycles, from the Jubilee of 457 BC to the Jubilee of AD 34, as you found in Lesson 2.

(Historians debate whether Isaiah's two years came from the calendar or from the war itself, so this is a checkpoint worth testing, not a proof.)

Practice Problems

Use the four CJC steps on each year:

- **26.** AD 2000
- **27.** AD 1776 (the year of the U.S. Declaration of Independence, when the Liberty Bell was already hanging in Philadelphia)
- **28.** AD 1492
- **29.** AD 1 **and** 1 BC. How far apart are they on the count? Why?
- **30.** AD 2050
- **31.** The year you were born

Bonus Lesson: What Day of the Week?

The CJC tells you where a **year** sits in the ancient count. But what about the **day of the week**? In 1882 a German mathematician named **Christian Zeller** published a formula that finds it for any date. It uses the same skills you've been practicing: whole-number division and remainders.

Get your numbers ready

- **q** = the day of the month
- **m** = the month number, with a twist: **March = 3, April = 4, ... December = 12**, and **January = 13, February = 14**, *counted as months of the year before*. (So for January 1, 2000, use m = 13 and the year 1999.)
- **K** = the **last two digits** of the year (for 2046, K = 46)
- **J** = the **first two digits** of the year (for 2046, J = 20)

In the formula, "÷" means whole-number division: **drop the remainder**.

The formula (for dates in the Gregorian calendar, 1582 and later)

$$\text{Total} = q + (13 \times (m + 1) \div 5) + K + (K \div 4) + (J \div 4) + (5 \times J)$$

Divide the **Total** by 7. The **remainder** is the day:

Remainder	0	1	2	3	4	5	6
Day	Saturday	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday

Worked example: July 4, 2046 (twenty years from now)

$$q = 4, m = 7, K = 46, J = 20$$

- $13 \times 8 = 104$, and $104 \div 5 = \mathbf{20}$ (drop the remainder)
- $K \div 4 = 46 \div 4 = \mathbf{11}$
- $J \div 4 = 20 \div 4 = \mathbf{5}$
- $5 \times J = \mathbf{100}$
- **Total** = $4 + 20 + 46 + 11 + 5 + 100 = \mathbf{186}$
- $186 \div 7 = 26$, remainder **4** → **Wednesday**

Worked example: July 4, 1776

$$q = 4, m = 7, K = 76, J = 17$$

- $13 \times 8 \div 5 = \mathbf{20}$; $76 \div 4 = \mathbf{19}$; $17 \div 4 = \mathbf{4}$; $5 \times 17 = \mathbf{85}$
- **Total** = $4 + 20 + 76 + 19 + 4 + 85 = \mathbf{208}$
- $208 \div 7 = 29$, remainder **5** → **Thursday**

The Declaration of Independence was adopted on a Thursday.

Going back 2,000 years: the Julian calendar

Before 1582, Europe used the **Julian calendar**, so older dates need a slightly different last part:

$$\text{Total} = q + (13 \times (m + 1) \div 5) + K + (K \div 4) + 5 + (6 \times J)$$

Worked example: April 7, AD 30

For the year 30, $K = 30$ and $J = 0$.

- $q = 7$, $m = 4$
- $13 \times 5 = 65$, and $65 \div 5 = \mathbf{13}$
- $K \div 4 = 30 \div 4 = \mathbf{7}$
- plus **5**, plus $6 \times 0 = \mathbf{0}$
- **Total** = $7 + 13 + 30 + 7 + 5 + 0 = \mathbf{62}$
- $62 \div 7 = 8$, remainder **6** → **Friday**

That confirms, with your own arithmetic, the Friday date used in Lessons 3 and 5.

Try it yourself

- **32.** What day of the week will **your birthday** fall on in **2046**?
- **33. December 25, 2026** (Gregorian)
- **34. January 1, 2000** (Gregorian; remember the January twist!)
- **35.** Check Lesson 3's answers with the **Julian** formula: **May 18, AD 30** and **May 28, AD 30**.

Answer Key

Lesson 1

- **1. AD 2021:** $B = 3,458 \rightarrow 3,457 \div 49 = 70 \text{ r } 27 \rightarrow \text{Cycle } 71 \rightarrow Y = 3,458 - 3,430 = 28 \rightarrow 28 \div 7 = 4 \text{ r } \mathbf{0} \rightarrow \text{rest year. } \checkmark$ It matches the Jewish calendar.
- **2. AD 2028:** $B = 3,465 \rightarrow \text{Cycle } 71 \rightarrow Y = 35 \rightarrow 35 \div 7 = 5 \text{ r } 0 \rightarrow \text{rest year.}$
- **3. AD 2043:** $B = 3,480 \rightarrow 3,479 \div 49 = 71 \text{ r } 0 \rightarrow \text{Cycle } 72 \rightarrow Y = 3,480 - 3,479 = \mathbf{1} \rightarrow \text{Jubilee.}$

Lesson 2

- **4.** $538 - 483 = \mathbf{55 \text{ BC}}$ (still BC, so no extra step).
- **5.** $520 - 483 = \mathbf{37 \text{ BC.}}$

- **6.** $483 - 444 = 39$, then $+1 = \text{AD } 40$. Only the 457 BC starting point reaches AD 27.
- **7.** The final week covers **AD 27, 28, 29, 30, 31, 32, and 33** (Positions 1 through 7). Its middle, Position 4, is **AD 30**.

Lesson 3

- **8. AD 30:** $B = 1,467 \rightarrow 1,466 \div 49 = 29 \text{ r } 45 \rightarrow \text{Cycle } 30 \rightarrow Y = 1,467 - 1,421 = 46 \rightarrow 46 \div 7 = 6 \text{ r } 4 \rightarrow \text{Position } 4$.
- **9. Day 40:** April 9–30 is 22 days, so day 40 is **May 18, a Thursday**.
- **10. Day 50: May 28, a Sunday**.
- **11.** $30 + 40 = \text{AD } 70$.

Lesson 4

- **12. 702 BC:** $B = 736 \rightarrow 735 \div 49 = 15 \text{ r } 0 \rightarrow \text{Cycle } 16 \rightarrow Y = 736 - 735 = 1 \rightarrow \text{Jubilee}$.
- **13. 605 BC:** $B = 833 \rightarrow 832 \div 49 = 16 \text{ r } 48 \rightarrow \text{Cycle } 17 \rightarrow Y = 833 - 784 = 49 \rightarrow 49 \div 7 = 7 \text{ r } 0 \rightarrow \text{rest year}$ (the last one before a Jubilee).
- **14.** $605 - 70 = 535 \text{ BC} \rightarrow B = 903 \rightarrow \text{Cycle } 19 \rightarrow Y = 21 \rightarrow \text{rest year}$.
- **15. 586 BC:** $Y = 19$, Position 5, an ordinary year. **538 BC:** $Y = 18$, Position 4, an ordinary year.

Lesson 5

- **16. 164 BC:** $B = 1,274 \rightarrow 1,273 \div 49 = 25 \text{ r } 48 \rightarrow \text{Cycle } 26 \rightarrow Y = 1,274 - 1,225 = 49 \rightarrow \text{rest year. } \checkmark$
- **17. 38 BC:** $B = 1,400 \rightarrow 1,399 \div 49 = 28 \text{ r } 27 \rightarrow \text{Cycle } 29 \rightarrow Y = 1,400 - 1,372 = 28 \rightarrow \text{rest year. } \checkmark$
- **18. AD 358:** $B = 1,795 \rightarrow 1,794 \div 49 = 36 \text{ r } 30 \rightarrow \text{Cycle } 37 \rightarrow Y = 1,795 - 1,764 = 31 \rightarrow 31 \div 7 = 4 \text{ r } 3 \rightarrow \text{Position } 3. \checkmark$
- **19.** $1/7$, about 14%. A formula that guessed would match the rhythm 1 time in 7.
- **20. April 3, AD 33 (Julian):** $q = 3, m = 4, K = 33, J = 0 \rightarrow 3 + 13 + 33 + 8 + 5 + 0 = 62 \rightarrow 62 \div 7 = 8 \text{ r } 6 \rightarrow \text{Friday. AD 33: } B = 1,470 \rightarrow 1,469 \div 49 = 29 \text{ r } 48 \rightarrow \text{Cycle } 30 \rightarrow Y = 1,470 - 1,421 = 49 \rightarrow 49 \div 7 = 7 \text{ r } 0 \rightarrow \text{Position } 7$, the end of the week, not the middle.

Lesson 6

- **21.** $70 \times 49 = 3,430 \text{ years}$.
- **22.** $3,430 - 1,437 = 1,993$, then $+1 = \text{AD } 1994$.

- **23. AD 1994:** $B = 3,431 \rightarrow 3,430 \div 49 = 70 \text{ r } 0 \rightarrow \text{Cycle } 71 \rightarrow Y = 1 \rightarrow \text{Jubilee.}$
- **24.** $1994 - 49 = 1945 \rightarrow B = 3,382 \rightarrow 3,381 \div 49 = 69 \text{ r } 0 \rightarrow \text{Cycle } 70 \rightarrow Y = 1 \rightarrow \text{Jubilee.}$
- **25.** $538 + 1,260 = 1798.$ **AD 538:** $Y = 15,$ **Position 1.** **AD 1798:** $B = 3,235 \rightarrow 3,234 \div 49 = 66 \text{ r } 0 \rightarrow \text{Cycle } 67 \rightarrow Y = 1 \rightarrow \text{Jubilee.}$

Practice

- **26. AD 2000:** $Y = 7 \rightarrow \text{rest year.}$
- **27. AD 1776:** $Y = 28 \rightarrow \text{rest year.}$
- **28. AD 1492:** $Y = 38 \rightarrow \text{Position } 3.$
- **29. AD 1:** $B = 1,438, Y = 17, \text{Position } 3.$ **1 BC:** $B = 1,437, Y = 16, \text{Position } 2.$ They're only **1 apart** on the count, because there's no year zero between them.
- **30. AD 2050:** $Y = 8 \rightarrow \text{Position } 1 \text{ (the first year of a new week).}$
- **31.** Answers will vary. Check with a partner!

Bonus Lesson

- **32.** Answers will vary. Check with a partner, then with a calendar app!
- **33. December 25, 2026:** $q = 25, m = 12, K = 26, J = 20 \rightarrow 13 \times 13 \div 5 = 33; 26 \div 4 = 6; 20 \div 4 = 5; 5 \times 20 = 100 \rightarrow \text{Total} = 25 + 33 + 26 + 6 + 5 + 100 = 195 \rightarrow 195 \div 7 = 27 \text{ r } 6 \rightarrow \text{Friday.}$
- **34. January 1, 2000:** use $m = 13$ and the year **1999** $\rightarrow q = 1, K = 99, J = 19 \rightarrow 13 \times 14 \div 5 = 36; 99 \div 4 = 24; 19 \div 4 = 4; 5 \times 19 = 95 \rightarrow \text{Total} = 1 + 36 + 99 + 24 + 4 + 95 = 259 \rightarrow 259 \div 7 = 37 \text{ r } 0 \rightarrow \text{Saturday.}$
- **35.** Julian formula, year 30 ($K = 30, J = 0$). **May 18:** $18 + 15 + 30 + 7 + 5 + 0 = 75 \rightarrow 75 \div 7 = 10 \text{ r } 5 \rightarrow \text{Thursday } \checkmark.$ **May 28:** $28 + 15 + 30 + 7 + 5 + 0 = 85 \rightarrow 85 \div 7 = 12 \text{ r } 1 \rightarrow \text{Sunday } \checkmark.$ Both match Lesson 3.

Formula Cards

The CJC formula (which year in the ancient count)

Step	What to do
1. Find B	BC: $B = 1438 - \text{year}.$ AD: $B = \text{year} + 1437$
2. Find the Cycle	$(B - 1) \div 49,$ drop the remainder, + 1
3. Find Y	$Y = B - (\text{Cycle} - 1) \times 49$ (Y is always 1 to 49)

Step	What to do
4. Find the Position	$Y \div 7$: remainder 1–6 = the position; remainder 0 = a rest year
Special	$Y = 1 \rightarrow$ a Jubilee
Year-zero rule	When counting across from BC to AD, subtract, then add 1

Zeller's formula (which day of the week)

Part	What to do
Numbers	q = day; m = month (March = 3 ... December = 12; January = 13, February = 14 of the year before); K = last two digits of the year; J = first two digits
Gregorian (1582 and later)	Total = $q + (13(m + 1) \div 5) + K + (K \div 4) + (J \div 4) + 5J$
Julian (before 1582)	Total = $q + (13(m + 1) \div 5) + K + (K \div 4) + 5 + 6J$
Answer	Total $\div 7$, and the remainder is the day: 0 = Saturday, 1 = Sunday, 2 = Monday, 3 = Tuesday, 4 = Wednesday, 5 = Thursday, 6 = Friday
Division	" $\div$ " always means drop the remainder, except the final step, where the remainder is the answer

Glossary

- **Accurate:** matching what is actually true or what actually happened.
- **AD / BC:** "Anno Domini" and "before Christ," the system most of the world uses to number years. There is no year zero.
- **Assyria / Babylon / Persia:** ancient empires of the Middle East.
- **Captivity:** a time when a people are taken away from their homeland.
- **Cycle:** 49 years (7 weeks of years), from one Jubilee to the next.
- **Day-for-a-year rule:** a rule stated in ancient Hebrew texts (Numbers 14:34; Ezekiel 4:6) for reading prophecies, in which each day stands for one year.
- **Generation:** in the Hebrew Bible, often counted as forty years.
- **Gospels:** four ancient accounts of the life of Jesus of Nazareth.
- **Gregorian calendar:** the calendar most of the world uses today, introduced in 1582.
- **Hebrew Bible:** the ancient collection of Jewish scriptures, which includes Leviticus, Numbers, Isaiah, Jeremiah, Ezekiel, and Daniel.
- **Hypothesis:** a testable statement about what you expect to be true.
- **Inclusive counting:** counting where the starting day (or year) is number 1.

- **Josephus:** a Jewish historian who wrote in the first century AD.
- **Jubilee:** the year that begins each 49-year cycle, a year of freedom and return.
- **Julian calendar:** the calendar Europe used before 1582, introduced by Julius Caesar in 46 BC.
- **Maccabees:** Jewish leaders who fought a war in the 160s BC, described in the ancient history 1 Maccabees.
- **Pentecost:** an ancient harvest festival, counted fifty days after the Passover season.
- **Position:** a year's place (1 to 7) in its week of years.
- **Precise:** giving exactly the same result every time.
- **Rest year (sabbatical year, shemitah):** every seventh year, when the land was not farmed.
- **Temple:** the central house of worship in ancient Jerusalem, destroyed by Rome in AD 70.
- **Week of years:** seven years, counted like the days of a week.
- **Zeller's formula:** a formula published by Christian Zeller in 1882 that finds the day of the week for any date.

A Note for Teachers

These lessons use ancient texts and traditional historical dates as **data**, the same way a history lesson uses primary sources. They don't ask students to accept any religious belief. All quotations of ancient texts are from the King James Version (1611), a historical English translation.

The CJC formula is exact arithmetic, and every example can be checked by hand. The starting year (1437 BC) and the event dates come from the Continuous Jubilee Calendar framework. Lesson 5 shows how its seven-year rhythm can be tested against independent records, and it states honestly how strong that test is. Zeller's formula (1882) is a standard method for finding the day of the week, and students can check every answer against a calendar app. For further reading, the full CJC papers are available at <https://greysmokez.github.io/CJC-Website/>.